

Contact:

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Markus Gokan

Mechanical Engineer, Eagle Scout

[My Portfolio](#)

Education

University of California San Diego, La Jolla, CA

Mechanical Engineering, Bachelors of Science, GPA 3.24

Sep 2021 - Jun 2025

Mechanical Engineering, Master of Science: Fluids and Mechanics, GPA 3.5

Sep 2025 - Jun 2026

Skills: Autodesk Fusion (Autodesk Certified), AutoCAD (Autodesk Certified), Solidworks, Inventor, Matlab, Manual Mill and Lathe, CNC, Laser Cutting, Microsoldering, Additive Manufacturing, Rapid Prototyping, Designing for Manufacturability, ROS2

Professional Experience

Troll Systems, Valencia, CA

Jul 2025 - Sep. 2025

Intern, Engineering and Systems Integration

- Designed test fixture for deployable aircraft antenna to simulate full system weight at early stages of manufacturing for Quality Department's testing
- Met real-time documentation update goals for assembly drawings, schematics, and part revisions
- Conducted MIL STD 461G and DO 160 qualification testing on as-built units for certification
- Characterized Radio Frequency patterns for antennas, insertion and return losses, INSS and Potentiometer tuning to ensure full system integration
- Wrote Acceptance Testing Procedures to meet exacting customer requirements

Kautex- Textron, Avilla, IN

Jun 2023 - Sep 2023

Intern, Manufacturing Engineering

- Designed 'Poka-Yoke' mechanism to validate force output of quality control devices, preventing a previous failure that had previously resulted in tens of thousands of defective products
- Reconfigured robotic vision software with a global zero, eliminating a tolerance stacking issue responsible for consistent false negative results from the new automated quality control
- Created an application to generate an automatic PARETO analysis of downtime caused by out-of-stock spare parts. Implementation of this program would ensure that 80% of parts-related downtime from the past 24 months would be avoided with just a 1% increase in spare parts inventory value.

Flow-Turn Inc, Union, NJ

Jun 2022 - Sep 2022

Intern, R&D and Engineering

- Built and tested sprocket driven conveyor-merge prototypes, seeking to replace traditional friction-driven designs with known reliability and controllability issues
- Pushed company full vertical integration goal forward by coordinating the workflow and nesting procedure of an AMADA fiber-optic laser cutting machine, reducing overhead and shortening project lead time from days to hours.
- Assembled one-off orders for customers who required specialty sensing capabilities or other features outside of the standard manufacturing processes

Project Teams

Suspension Subteam: Triton Baja SAE

- Designing double wishbone suspension for the first buggy that Triton Baja has produced. Validated suspension/steering geometry and conducted dynamic and FEA analysis on parts and assemblies

Technical Projects

Wave-Enabled Energy Generation Device (Patent Pending):

- Presented at ASEE, Cal-Poly Pomona, April 12th 2025 and ASME, San Diego, April 25th 2025
- Optimized fluid distribution and medium pressure seal of a modular 3D printed manifold.
- Developed a test fixture with real-time data acquisition for evaluating multi-channel configurations and changing flow rates to optimize microchannel geometry and performance.
- Achieved nano-watts of power generation per unit, three orders of magnitude greater than existing literature had proved possible, at a fraction of the cost and manufacturing time

MAE 148 Autonomous Vehicles: El Cochito:

- Designed and programmed a self-driving RC car powered by a VESC, NVidia Jetson Nano, OAK-D RGBD camera, LD06 lidar, and GNSS board.
- Using deep learning, PID tuning, and a ROS2 AI model we achieved autonomous lane tracking, obstacle recognition, and point to path navigation then integrated a pose estimation model capable of responding to various student postures such as raising their hand, standing, and sleeping